

Experiment 2: Circle Detection using Hough Transform

Aim:

To detect circular objects in an image using the Hough Circle Transform technique.

Theory:

The Hough Circle Transform is a feature extraction method used in image processing to detect circular shapes. It works by transforming edge-detected pixels into a parameter space and identifying potential circle centers based on votes. Before applying the transform, the image is converted to grayscale, blurred to remove noise, and edges are detected using the Canny method. The OpenCV function `cv2.HoughCircles()` automates this process by taking parameters such as resolution, minimum distance, and thresholds to locate and validate circular patterns in the image.

Algorithm:

- Start
- Import required libraries (`cv2`, `numpy`)
- Load the input image using `cv2.imread()`
- Convert the image to grayscale using `cv2.cvtColor()`
- Apply Gaussian blur to remove noise
- Apply Canny edge detection to detect edges
- Use `cv2.HoughCircles()` to detect circular shapes in the blurred image
- If circles are detected, draw them on a copy of the original image
- Display the original image, edges, and detected circles
- Release resources and close all OpenCV windows
- End

Flowchart:

Start → Import Libraries → Load Image → Convert to Grayscale → Apply Gaussian Blur → Canny Edge Detection → Hough Circle Transform → Draw Circles → Display Output → End

Viva Questions:

- What are the parameters used in `cv2.HoughCircles()`?
- Why is Gaussian blur applied before edge detection?
- How does Hough Circle Transform work?